

REVIEW ARTICLE

Clinical Implications of Minipuberty

Anders Juul, M.D., D.M.Sc., Ph.D.¹⁻³

SUMMARY

Minipuberty refers to a specific period in early postnatal life with high activity of the hypothalamic–pituitary–gonadal (HPG) hormone axis. In infant boys 1 to 3 months of age, high concentrations of follicle-stimulating hormone and luteinizing hormone are released, which results in high levels of gonadal hormones from testicular Leydig cells (testosterone and insulin-like 3) and Sertoli cells (inhibin B and antimüllerian hormone). The HPG axis is also active in infant girls, who have adult levels of follicle-stimulating hormone and luteinizing hormone during this period. Immediately after minipuberty, the HPG axis is silenced for approximately 10 years and is reactivated only with the onset of puberty. Thus, minipuberty represents an early window for diagnosing disorders of sexual differentiation and rare endocrine disorders, such as congenital hypogonadotropic hypogonadism, and may help guide dosing of therapeutic interventions including gonadotropin therapy. Of note, minipuberty predicts adult reproductive capacity.

Author affiliations are listed at the end of the article. Anders Juul can be contacted at anders.juul@regionh.dk or at the Department of Growth and Reproduction, Rigshospitalet, Section 5064, Blegdamsvej 9, DK-2100 Copenhagen, Denmark.

N Engl J Med 2026;394:1204-14.

DOI: 10.1056/NEJMra2305989

Copyright © 2026 Massachusetts Medical Society.

CME



PUBERTY ENCOMPASSES THE COMPLEX TRANSITION FROM CHILDHOOD TO reproductive adulthood. In humans, puberty occurs after a hormonally quiescent childhood that is preceded by a phase in infancy known as minipuberty. In male infants, luteinizing hormone (LH) and testosterone during minipuberty are present at levels similar to those in adults but subsequently become undetectable. This phenomenon was shown in infant boys more than 50 years ago. The physiological role of minipuberty is not clear, but it has been suggested that this period presents a narrow window of opportunity for diagnosing hypogonadal conditions in male infants.¹ In female infants, levels of follicle-stimulating hormone (FSH) fluctuate during minipuberty and correspond to levels observed during the menstrual cycle in women and girls,² and estradiol levels are similar to those observed in women and girls of reproductive age.³ This review summarizes our current understanding of minipuberty and describes its potential clinical implications.

PHYSIOLOGY OF PUBERTY

Human puberty is controlled by the brain, where integration of multiple neuronal inputs initiates secretion of gonadotropin-releasing hormone (GnRH) by the hypothalamic GnRH pulse generator.⁴ Kisspeptin, neurokinin B, and dynorphin neurons stimulate GnRH neurons, whereas makorin ring finger protein 3 (MKRN3) inhibits GnRH secretion and is considered to be a brake on puberty.⁵ Loss-of-function variants in MKRN3 release this brake and result in central precocious puberty.⁶

GnRH is released in a pulsatile manner to the anterior pituitary lobe through portal vessels and subsequently stimulates pituitary gonadotrophic cells to secrete FSH and LH into the peripheral circulation. In girls, LH stimulates ovarian theca

cells to produce androgens, which enter adjacent granulosa cells; FSH-induced aromatization in these cells results in estradiol production. In boys, LH stimulates testicular Leydig cells to produce and release testosterone and insulin-like 3 (INSL3).⁵⁻⁷ The activation of the hypothalamic–pituitary–gonadal (HPG) axis in puberty is preceded by two earlier periods of hormonal activity, the first during fetal life and the second during infancy (minipuberty). Elucidating the complex mechanisms that result in puberty and contribute to adult reproductive health requires a more detailed physiological understanding of these three periods of HPG activation than we currently have. Specifically, minipuberty may have particular clinical relevance for later reproductive function.

FIRST SURGE IN SEX HORMONES IN UTERO

Because maternal pituitary hormones do not cross the placenta, the fetus must secrete its own pituitary hormones. In the fetus, GnRH neurons migrate with olfactory neurons from the nasal placode to the hypothalamus and can be detected there at approximately 14 to 15 weeks' gestation. FSH and LH can be detected at 12 to 14 weeks' gestation in the pituitary gland.^{7,8}

In male fetuses, testicular Leydig cells start producing testosterone at 7 to 8 weeks' gestation. Before 14 to 15 weeks' gestation, Leydig cells produce testosterone independently of LH stimulation but are dependent on stimulation by placental human chorionic gonadotropin (hCG). Placental hCG thus supports androgen action during the “masculinization programming window,” between 8 and 14 weeks' gestation, when the penis is formed and urethral closure is completed. Starting at 15 to 16 weeks' gestation, increasing GnRH and downstream pulsatile release of LH increase testosterone production by Leydig cells, stimulating penile growth. Leydig cells also produce INSL3, which is responsible for the transabdominal phase of testicular descent. In midgestation, high placental estrogen levels exert negative central feedback and decrease gonadotropin and testosterone levels until delivery (Fig. 1).

Fetal testosterone in the third trimester is important for penile growth and the inguinoscrotal

phase of testicular descent. Boys with congenital hypogonadotropic hypogonadism have low fetal testosterone levels in the second and third trimesters and typically show signs of undervirilization at birth, including micropenis (stretched penile length, <2.5 cm or ≥ 2 SD below the population mean) and cryptorchidism (the absence of one or both testes in the scrotum). In the largest study of male term neonates (who were predominantly White), the mean ($\pm$ SD) penile length was 3.5 ± 0.4 cm.⁹ Penile length can depend on room temperature and the degree of stretching¹⁰ and can vary according to ethnic group.^{11,12} In addition to micropenis and cryptorchidism, other signs of fetal undervirilization in newborn boys include hypospadias, a short anogenital distance, and a low external genitalia score.¹³

The anogenital distance is considered a postnatal readout of androgen action during the masculinization programming window¹⁴ and shows strong sexual dimorphism, which helps distinguish male from female phenotype in the third trimester (by means of ultrasonography)¹⁵ and at birth.¹⁶ In the postnatal period, the short anogenital distance in boys born with cryptorchidism, hypospadias, or anorchia, as compared with the distance in normogonadal boys,^{17,18} represents insufficient fetal androgen action. The anogenital distance during minipuberty correlates with the anogenital distance at 9 years of age in both boys and girls,¹⁹ which suggests that this measurement in utero has implications for the rest of life.

The external genitalia score¹³ grades five anatomical genital landmarks (the degree of labio-scrotal fusion, length of the genital tubercle, position of the urethral meatus, and locations of the right and left gonads) on a scale of 0 to 12. Both the anogenital distance and the external genitalia score may reflect intrauterine androgen action during minipuberty.

SECOND SURGE OF SEX HORMONES IN INFANCY

A second surge of sex hormones in infancy leads to myriad physiological changes in both sexes. Levels of sex hormones released during this period of minipuberty are significantly correlated with growth velocity in the first 6 months of life.²⁰

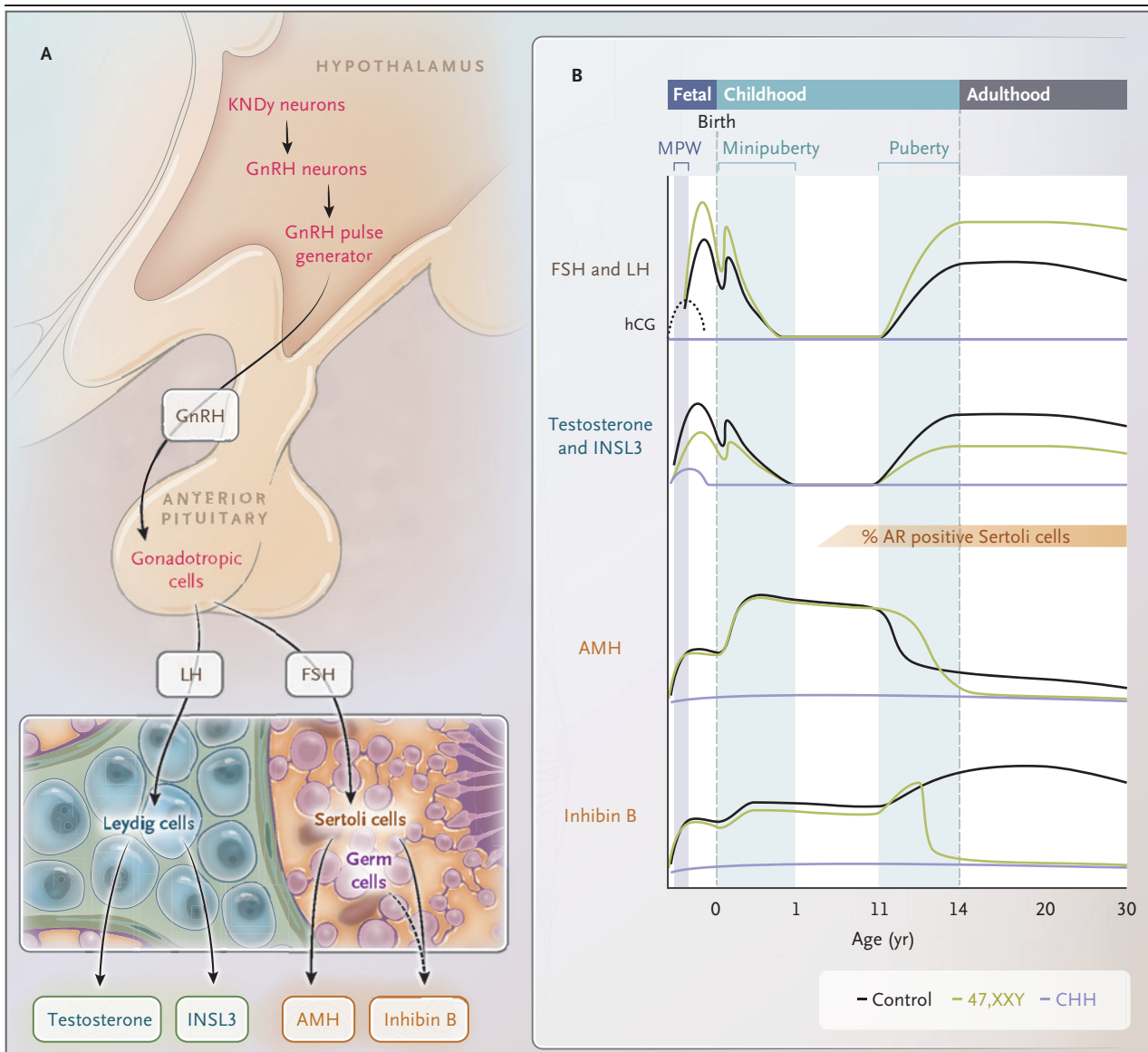


Figure 1. The Hypothalamic–Pituitary–Gonadal (HPG) Hormone Axis in Male Infants, Children, and Adolescents.

The HPG axis is activated by the integration of neuronal inputs from kisspeptin, neurokinin B, and dynorphin (KNDy) neurons, which stimulate hypothalamic gonadotropin-releasing hormone (GnRH) neurons to induce a pulsatile release of GnRH, which in turn stimulates gonadotropic cells in the anterior lobe of the pituitary gland to secrete follicle-stimulating hormone (FSH) and luteinizing hormone (LH) (Panel A). FSH stimulates Sertoli cells to produce inhibin B and antimüllerian hormone (AMH). Circulating inhibin B consists of two subunits (α and β); in prepubertal boys, both subunits are derived from Sertoli cells in the testes, whereas in pubertal boys and adult men, the β subunit of inhibin B is derived from germ cells (spermatocytes and round spermatids). LH stimulates Leydig cells to produce testosterone and insulin-like 3 (INSL3). Gonadotropin secretion undergoes three distinct surges: in utero, in early infancy (minipuberty), and during puberty (Panel B). In early pregnancy, placental human chorionic gonadotropin (hCG) stimulates LH receptors on fetal Leydig cells to produce testosterone in the masculinization programming window (MPW). Patients with congenital hypogonadotropic hypogonadism (CHH) have undetectable FSH and LH at all times, whereas patients with Klinefelter syndrome (the 47,XXY karyotype) have higher gonadotropin levels than controls. Increasing testosterone levels are seen in controls, as well as in patients with congenital hypogonadotropic hypogonadism or the 47,XXY karyotype. After birth, testosterone levels are markedly increased in controls, with peak levels at 1 month of age. At 6 to 12 months of age, testosterone becomes undetectable and remains so until puberty starts. The increase in testosterone and INSL3 levels is reduced in boys with the 47,XXY karyotype, and the levels of testosterone, INSL3, inhibin B, and AMH remain very low at all ages in patients with congenital hypogonadotropic hypogonadism. Androgen receptor (AR) expression in Sertoli cells begins to increase gradually at 4 years of age, and after puberty 100% of Sertoli cells express AR.

Growth deceleration occurs in infants with congenital hypogonadotropic hypogonadism who have low levels of sex hormones during minipuberty.²¹ In healthy infants, sex differences in body composition, postulated to be related to the effects of sex hormones during minipuberty, emerge in the first few months of life and include lower adiposity in boys than in girls.²² Negative correlations between testosterone and LH levels at 2 months of age and the body-mass index at up to 6 years of age, as well as positive correlations between estradiol levels and skin-fold thickness at up to 6 years of age in boys (but not girls), have also been reported.²³

The sex hormones released in minipuberty might also have neurobehavioral effects. High testosterone levels during minipuberty and penile growth between birth and minipuberty have been associated with male-type play behavior at 3 to 4 years of age^{24,25} and with less expressive language development than is typical in girls,^{26,27} which supports the role of testosterone in neurobehavioral sexual differentiation during infancy.

MINIPUBERTY IN INFANT BOYS

Gonadotropin and testosterone levels are low during the first few postnatal weeks. In classic studies, serum FSH and LH and plasma testosterone levels increased in male but not female infants from birth to the age of 1 to 3 months; the levels then declined to prepubertal levels at 7 months of age.^{28,29} Many subsequent cross-sectional studies have documented high levels of FSH, LH, testosterone, estradiol, inhibin B, antimüllerian hormone, and INSL3 during minipuberty³⁰; however, the exact ages when hormone levels peaked were unknown.

Longitudinal urine^{31,32} and serum³³⁻³⁵ testing in healthy boys has further delineated dynamic changes in the HPG axis according to age. Infant boys have two separate hormonal surges during minipuberty — an earlier one involving Leydig cells and a later one involving Sertoli-cell markers. A longitudinal study involving 119 healthy boys showed that FSH and LH levels peaked at 11 to 18 days after birth, testosterone and INSL3 levels (Leydig-cell markers) peaked at 1 month of age, and inhibin B and antimüllerian hormone levels (Sertoli-cell markers) peaked at 4 to 5 months of age, with concomitant increases in

testicular volume.³⁵ This increase in testosterone levels may promote final testicular descent, as evidenced by a decrease in the prevalence of cryptorchidism from 9% at birth to 2% at 3 months of age.³⁶ Testosterone and dihydrotestosterone levels correlate with penile length and growth,^{9,11} and inhibin B levels correlate with testicular volume during minipuberty.³⁷

The first postnatal surge in gonadotropins is associated with a proliferation of Sertoli cells,³⁸ key components in adult spermatogenesis, as well as with a subsequent increase in testicular volume.³⁵ The undifferentiated Sertoli cells do not express androgen receptors at this time, so spermatogenesis is delayed until more androgen receptors are expressed with age (Fig. 2).³⁹ Finally, the LH:FSH ratio increases during minipuberty and is significantly higher in boys than in girls.^{2,39}

At 3 to 6 months of age, levels of LH, testosterone, and INSL3 decline steeply and are largely undetectable from 12 months of age until puberty. Levels of FSH also decline but remain detectable. Conversely, levels of inhibin B and antimüllerian hormone remain high until the LH-induced testosterone surge during puberty leads to Sertoli-cell differentiation and antimüllerian hormone down-regulation.⁴⁰ Because of these dynamic changes in reproductive hormones during minipuberty, age-related reference ranges for sex hormones are needed, and correction for prematurity may be required.³¹

In a longitudinal study spanning 20 years that involved 259 healthy infant boys, testosterone levels during minipuberty were found to predict the sperm count at 18 to 20 years of age, and infantile penile length and testicular volume were correlated with penile length and testicular volume in adulthood.⁴¹ These findings are the strongest evidence to date that minipuberty may predict adult reproductive function in healthy men.

MINIPUBERTY IN INFANT GIRLS

The hormonal peaks during minipuberty are less distinct in girls than in boys. Ovarian activity has been shown during minipuberty,⁴² and a negative correlation between the FSH surge in minipuberty and ovarian follicle size and total follicle count has been documented on the basis of ultrasonography.^{43,44} Cross-sectional studies

have shown elevated urine or serum levels of FSH, LH, inhibin A and B, and estradiol during female minipuberty, although with a large degree of interindividual variation. A longitudinal study in infant girls showed a biphasic peak in reproductive hormone levels during minipuberty.⁴⁵ The approximate timing of the first peak was postnatal days 15 to 27, followed by a nadir in all hormones at postnatal days 58 to 92. The approximate timing of the second peak was postnatal days 107 to 125 for inhibin B, antimüllerian hormone, estrone, estradiol, and sex hormone-binding globulin and days 164 to 165 for LH and FSH.

During minipuberty, estradiol stimulates uterine growth and breast development^{46,47} in all infant girls.⁴⁷ The levels of reproductive hormones during female minipuberty may predict adult reproductive function,⁴⁸ as suggested by a large longitudinal study⁴⁹ showing that levels of antimüllerian hormone in minipuberty correlated with those in puberty and reflected the number

of ovarian follicles in adolescent girls. Measurement of antimüllerian hormone levels during female minipuberty may thus serve as a useful clinical tool for predicting future ovarian activity in healthy girls.⁴⁹

DIAGNOSTIC EVALUATION DURING MINIPUBERTY

The transient elevation of the HPG axis during minipuberty enables diagnostic evaluation of the entire hormonal axis. Infants with possible hypogonadism on physical examination may benefit from an evaluation during minipuberty.

Infants with primary gonadal failure (hypergonadotropic hypogonadism) due to bilateral cryptorchidism, anorchia (“vanishing testes”), Klinefelter syndrome (47,XXY), mixed gonadal dysgenesis (45,XO/46,XY), or Turner syndrome (45,XO) have elevated levels of gonadotropins with undetectable or low-normal levels of gonadal hormones during minipuberty.² In contrast, infants

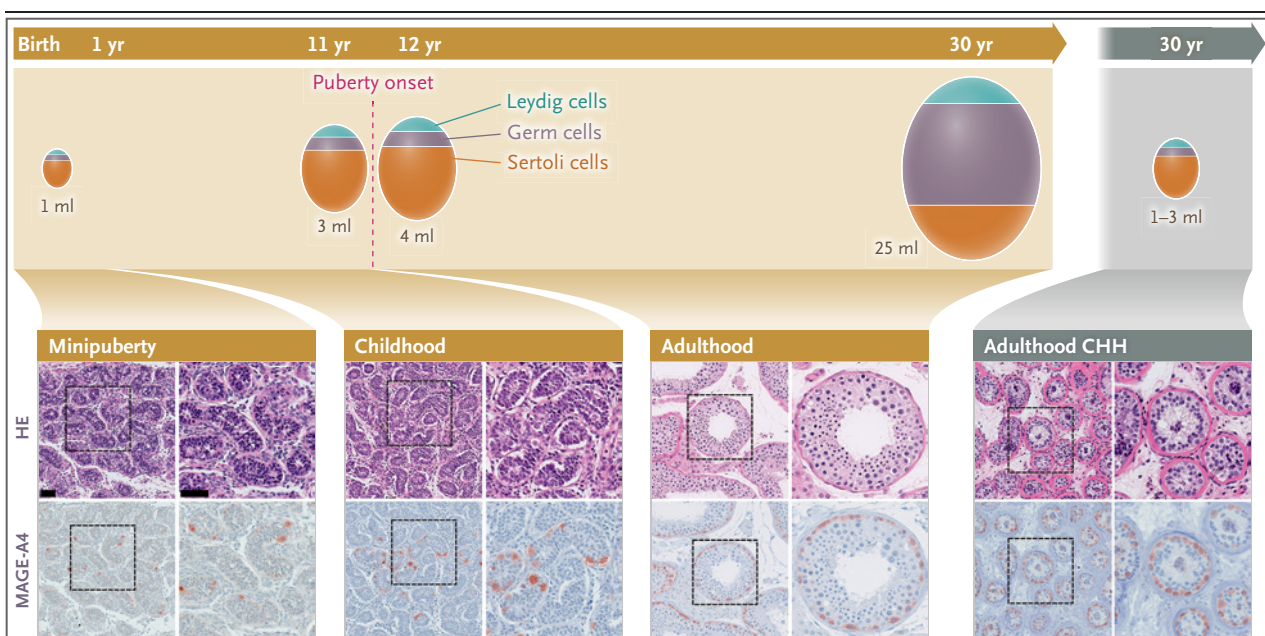


Figure 2. Testicular Volume, Composition, and Histologic Features According to Age.

Testicular volume and the proportions of seminiferous tubules, Sertoli cells, and Leydig cells are shown, as well as the histologic features of human testes during minipuberty, childhood, and adulthood (hematoxylin and eosin [HE] stain). Features of testes in an adult with congenital hypogonadotropic hypogonadism are shown for comparison. Germ cells are stained with MAGE-A4. The tubular diameter is increased in the adult control, whereas the tubular diameter remains small in adults with congenital hypogonadotropic hypogonadism.

with congenital GnRH deficiency (hypogonadotropic hypogonadism) have undetectable or subnormal levels of both gonadotropins and gonadal hormones during this period.

HYPOGONADOTROPIC HYPOGONADISM IN MINIPUBERTY

Congenital hypogonadotropic hypogonadism has a prevalence of 1 case per 10,000 persons⁴ and can be either isolated or part of multiple pituitary hormone deficiencies. Children with this condition do not have the three standard surges in HPG hormones during fetal life, minipuberty, and puberty. Although boys with congenital hypogonadotropic hypogonadism typically have undervirilization at birth and during minipuberty, the condition is still frequently underdiagnosed. Since infants with congenital hypogonadotropic hypogonadism are not testosterone-deficient in early pregnancy, when placental hCG is the main source of testosterone production, they have urethral closure and do not have hypospadias. However, testosterone deficiency in later pregnancy and minipuberty results in micropenis and cryptorchidism. Unfortunately, most cases of congenital hypogonadotropic hypogonadism are diagnosed only after a presentation with delayed puberty or infertility. Girls with the condition have no clinical manifestations in infancy or childhood. Congenital hypogonadotropic hypogonadism can be diagnosed on the basis of subnormal FSH and LH levels during minipuberty, in addition to low or undetectable levels of testosterone, INSL3 (a measurement that is not routinely available in most laboratories), antimüllerian hormone, and inhibin B.^{50,51} Diagnostic evaluation during this window is recommended for male newborns with undervirilization who are suspected to have hypogonadism.⁵²

HYPERGONADOTROPIC HYPOGONADISM IN MINIPUBERTY

Boys with nonpalpable testes may have either anorchia or bilateral cryptorchidism with intraabdominal testes. Traditionally, such patients may undergo hCG testing, magnetic resonance imaging, or laparoscopy to distinguish between the two conditions. However, these conditions

can also be distinguished through biochemical testing during minipuberty. Anorchia results in undetectable testosterone, INSL3, antimüllerian hormone, and inhibin B levels and results in elevations in FSH and LH levels that are higher than those in patients with cryptorchidism with intraabdominal testes.^{53,54}

Klinefelter syndrome (47,XXY) is the most prevalent sex-chromosome disorder (occurring in 1 in 660 births) but is frequently overlooked because of heterogeneous phenotypes. It is estimated that the disorder is never diagnosed in 50% of boys with a 47,XXY karyotype. Biochemical signs of primary hypogonadism, including elevated or high-normal FSH and LH levels⁵⁵ and low-normal testosterone levels,⁵⁶ may be seen during minipuberty. However, these biochemical abnormalities are not always present.

Mixed gonadal dysgenesis is characterized by a mosaic 45,XO/46,XY karyotype. Most infants with mixed gonadal dysgenesis are phenotypically male and have an LH:FSH ratio and testosterone, inhibin B, and antimüllerian hormone levels that are within normal reference ranges for boys during minipuberty.² Measurement of these hormones during minipuberty may inform physicians and families in cases of difficult sex assignment. Specifically, in one study, the LH:FSH ratio in 15 infants with the 45,XO/46,XY karyotype was found to correspond to the eventual assigned sex, whereas the external genitalia score was less discriminative.³⁹

Patients with mosaic or nonmosaic Turner syndrome (45,XO karyotype) typically receive the diagnosis in late childhood or adolescence, after presenting with growth failure, delayed puberty, or primary amenorrhea. During minipuberty, girls with Turner syndrome have FSH and LH levels that are markedly above reference levels, but LH:FSH ratios are within the normal range for girls.² Girls with Turner syndrome have the typical biphasic peak in hormone levels, with an exaggerated peak in gonadotropin levels at puberty.⁵⁷ Most patients with nonmosaic Turner syndrome have undetectable levels of inhibin B and antimüllerian hormone.²

Variants in AR, the gene encoding androgen receptor, may lead to a complete or partial androgen insensitivity syndrome, with a phenotype ranging from female to male but with

undervirilization and gynecomastia during puberty. During minipuberty, testosterone and LH levels are normal in infant boys with partial androgen insensitivity syndrome but are low in girls with the complete syndrome.⁵⁸ However, such girls have LH:FSH ratios, antimüllerian hormone levels, and inhibin B levels that are within the normal ranges for boys.²

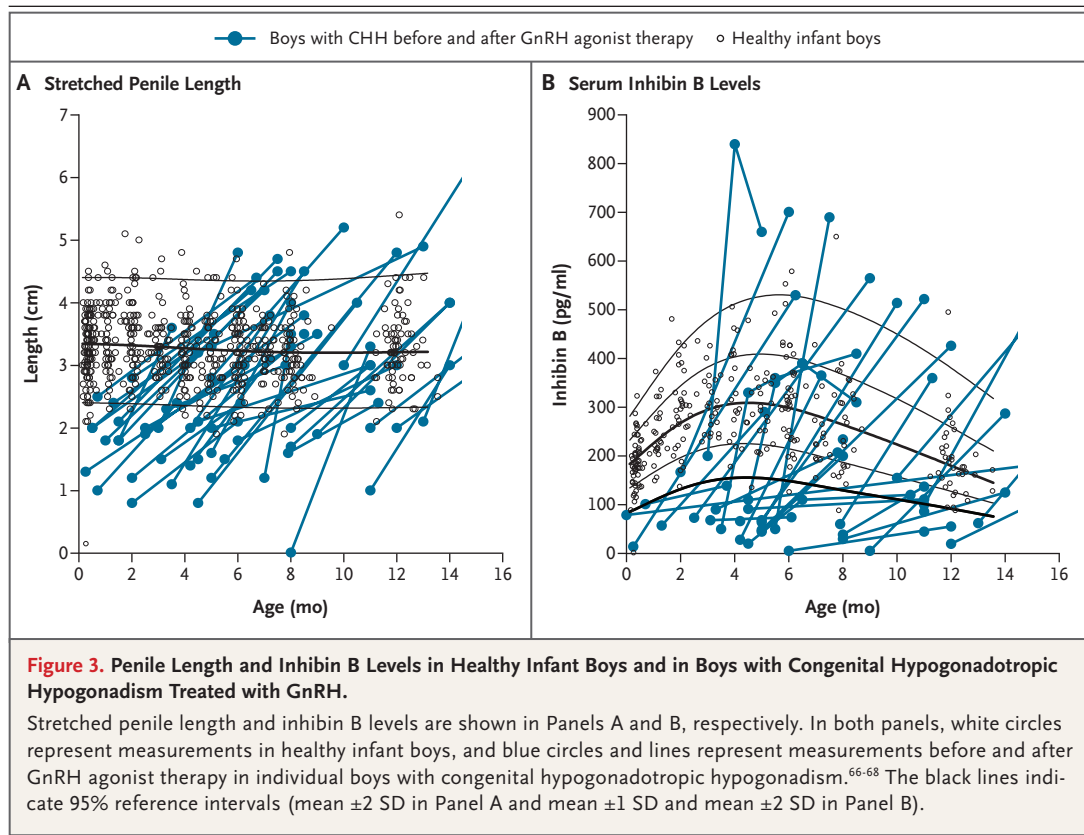
THERAPEUTIC INTERVENTIONS IN INFANTS WITH DISORDERED MINIPUBERTY

Hormonal imbalances in infants with endocrinopathies can be corrected during minipuberty, with potentially enduring positive effects on adult reproductive outcomes. Appropriate management of congenital hypogonadotropic hypogonadism in newborns remains a major clinical challenge. For boys with undervirilization, whether treatment should consist of traditional androgen supplementation alone or in combination with gonadotropins to mimic normal physiology is unclear.

TESTOSTERONE TREATMENT

In infants with disordered minipuberty, restoring reproductive hormone levels to physiologic levels may be important. In observational studies, testosterone injections or topical dihydrotestosterone gel administered in infants with severe micropenis have been associated with normal adult penile length.⁵⁹

Testosterone treatment has also emerged as off-label therapy for infant boys with Klinefelter syndrome.⁶⁰ In a retrospective study, testosterone injections during minipuberty in boys with Klinefelter syndrome were associated with improved neurocognitive functioning.⁶⁰ However, these findings were not replicated in subsequent randomized trials, and testosterone use for this purpose is not recommended. Favorable effects of short-term testosterone treatment on lean mass, growth velocity, and penile length were shown in a recent randomized, placebo-controlled trial involving 71 infants with Klinefelter syndrome.⁶¹ The intervention did not yield measurable beneficial effects on neurodevelopment, and potential adverse effects on the endogenous HPG axis were



reported. Since not all infant boys with Klinefelter syndrome have subnormal testosterone levels during minipuberty, it is not clear which boys should be treated with testosterone.

GONADOTROPIN OR GNRH THERAPY

Boys with congenital hypogonadotropic hypogonadism have pubertal failure and infertility if the disorder is left untreated. Current management includes administration of testosterone or dihydrotestosterone during minipuberty (to increase penile size) and surgical orchidopexy (for undescended testes). In early adolescence, testosterone treatment is used for pubertal development, and gonadotropin therapy is used to stimulate spermatogenesis.⁶² Several years of such therapy may be necessary to induce spermatogenesis, and results depend on testicular size, inhibin B levels, and the absence or presence of previous pubertal development.⁶³ Induction of spermatogenesis with gonadotropins is much more rapid and results in higher sperm counts in patients with adult-onset hypogonadotropic hypogonadism (who have previously undergone minipuberty and puberty) than in patients with congenital hypogonadotropic hypogonadism (who do not undergo minipuberty or puberty).

A recent advance is the use of gonadotropin-replacement therapy in boys with congenital hypogonadotropic hypogonadism during minipuberty and puberty⁶² to more closely mimic

physiologic hormonal changes and potentially promote successful spermatogenesis induction in adulthood. This approach may increase long-term fertility in boys with congenital hypogonadotropic hypogonadism but requires further evaluation.

A systematic review⁶⁴ involving 11 studies and two recent case series^{65,66} provided data on 88 infant boys with congenital hypogonadotropic hypogonadism treated with gonadotropins (Table S1 in the Supplementary Appendix, available with the full text of this article at NEJM.org). At the time that therapy was initiated, all the infants had micropenis and two thirds had bilateral cryptorchidism. The median age at initiation of therapy was 4 months, and the duration of treatment ranged from 2 to 7 months. A total of 36 infants received subcutaneous injections, and 35 received subcutaneous pump therapy. FSH priming (which is standard before gonadotropin therapy in pubertal or adult males with congenital hypogonadotropic hypogonadism) was not needed because Sertoli cells lack the androgen receptor in infancy.

In these studies, administration of gonadotropins during minipuberty, regardless of the route of administration, was associated with normal or supranormal testosterone, antimüllerian hormone, and inhibin B levels and increased penile length (mean, 1.8 cm before treatment and 3.7 cm after treatment) (Fig. 3). In addition, gonadotropin therapy was associated with testicular

KEY POINTS

CLINICAL IMPLICATIONS OF MINIPUBERTY

- The hypothalamic–pituitary–gonadal (HPG) hormone axis regulates gonadal function and has three hormonal surges (during fetal development, infancy, and puberty).
- The first postnatal activation of the HPG axis, which occurs 1 to 6 months after birth and is known as minipuberty, includes two distinct hormonal peaks in infant boys: Leydig-cell markers peak 1 to 2 months after birth, and Sertoli-cell markers peak 4 to 5 months after birth.
- Levels of sex hormones during minipuberty are similar to adult values, decrease at 6 to 12 months, and remain low for approximately the next 10 years, until the reawakening of the HPG axis during puberty.
- Levels of sex hormones during minipuberty may regulate penile growth in boys, as well as linear growth, body composition, and neurocognitive development, and they may predict adult reproductive function.
- Levels of antimüllerian hormone during minipuberty in infant girls predict the number of ovarian follicles in adolescent girls.
- Minipuberty offers a diagnostic window for identifying disorders of sexual development in male infants with undervirilization; early diagnosis is important, since these disorders are associated with lifelong sequelae if left untreated.
- Early replacement of gonadotropins in infant boys with congenital hypogonadotropic hypogonadism in order to mimic minipuberty is associated with increased testicular and penile size and may have long-term positive reproductive effects.

descent in 73% of the treated infants. One participant was evaluated 33 years later and showed marked testicular growth after only 12 months of adult gonadotropin therapy, with a total sperm count of 25 million spermatozoa. His wife conceived spontaneously,⁶⁷ more quickly than expected for a couple in which one partner has congenital hypogonadotropic hypogonadism.⁶⁸ Treatment with gonadotropins during minipuberty in patients with congenital hypogonadotropic hypogonadism may increase future fertility, but long-term follow-up is required.

CONCLUSIONS

Minipuberty offers an early diagnostic window for evaluating infants with suspected central or peripheral hypogonadism. Sex hormone levels during minipuberty appear to predict the reproductive outcome in adulthood. Consequently, the

early surge in sex hormones during minipuberty might be replicated by therapeutic interventions in hypogonadal infant boys. This precision-based approach to the diagnosis and management of disorders of sex development has the potential to alter health outcomes, particularly for boys with congenital hypogonadotropic hypogonadism, by ensuring timely and physiologic hormonal replacement for gonadal development and maturation.

Disclosure forms provided by the author are available with the full text of this article at NEJM.org.

Clement D. Lee, M.D., assisted with the editorial handling of this article for the *Journal*.

AUTHOR INFORMATION

¹Department of Growth and Reproduction, Copenhagen University Hospital—Rigshospitalet, Copenhagen; ²Department of Clinical Medicine, University of Copenhagen, Copenhagen; ³Research Center in Endocrine Disruption of Male Reproduction and Child Health, Rigshospitalet and University of Copenhagen, Copenhagen.

REFERENCES

1. Grumbach MM. A window of opportunity: the diagnosis of gonadotropin deficiency in the male infant. *J Clin Endocrinol Metab* 2005;90:3122-7.
2. Johannsen TH, Main KM, Ljubicic ML, et al. Sex differences in reproductive hormones during mini-puberty in infants with normal and disordered sex development. *J Clin Endocrinol Metab* 2018;103:3028-37.
3. Chellakooty M, Schmidt IM, Haavisto AM, et al. Inhibin A, inhibin B, follicle-stimulating hormone, luteinizing hormone, estradiol, and sex hormone-binding globulin levels in 473 healthy infant girls. *J Clin Endocrinol Metab* 2003;88:3515-20.
4. Boehm AP, Bouloux P-M, Dattani MT, et al. Expert consensus document: European consensus statement on congenital hypogonadotropic hypogonadism — pathogenesis, diagnosis and treatment. *Nat Rev Endocrinol* 2015;11:547-64.
5. Hughes IA. Releasing the brake on puberty. *N Engl J Med* 2013;368:2513-5.
6. Abreu AP, Dauber A, Macedo DB, et al. Central precocious puberty caused by mutations in the imprinted gene *MKRN3*. *N Engl J Med* 2013;368:2467-75.
7. Clements JA, Reyes FI, Winter JSD, Fairman C. Studies on human sexual development. III. Fetal pituitary and serum, and amniotic fluid concentrations of LH, CG, and FSH. *J Clin Endocrinol Metab* 1976;42:9-19.
8. Reyes FI, Boroditsky RS, Winter JSD, Fairman C. Studies on human sexual development. II. Fetal and maternal serum gonadotropin and sex steroid concentrations. *J Clin Endocrinol Metab* 1974;38:612-7.
9. Boas M, Boisen KA, Virtanen HE, et al. Postnatal penile length and growth rate correlate to serum testosterone levels: a longitudinal study of 1962 normal boys. *Eur J Endocrinol* 2006;154:125-9.
10. Vasudevan G, Manivarmane, Bhat BV, Bhatia BD, Kumar S. Genital standards for south Indian male newborns. *Indian J Pediatr* 1995;62:593-6.
11. Fok TF, Hon KL, So HK, et al. Normative data of penile length for term Chinese newborns. *Biol Neonate* 2005;87:242-5.
12. Jarrett OO, Ayoola OO, Jonsson B, Albertsson-Wikland K, Ritzen EM. Penile size in healthy Nigerian newborns: country-based reference values and international comparisons. *Acta Paediatr* 2014;103:442-6.
13. van der Straaten S, Springer A, Zecic A, et al. The External Genitalia Score (EGS): a European multicenter validation study. *J Clin Endocrinol Metab* 2020;105:dgz142.
14. van den Driesche S, Kolovos P, Platts S, Drake AJ, Sharpe RM. Inter-relationship between testicular dysgenesis and Leydig cell function in the masculinization programming window in the rat. *PLoS One* 2012;7(1):e30111.
15. Fischer MB, Mola G, Priskorn L, et al. Longitudinal evaluation of fetal and infant AGD in healthy children: association with penile size, testosterone, and DHT. *J Clin Endocrinol Metab* 2024;109:3087-95.
16. Fischer MB, Ljubicic ML, Hagen CP, et al. Anogenital distance in healthy infants: method-, age- and sex-related reference ranges. *J Clin Endocrinol Metab* 2020;105:2996-3004.
17. Thankamony A, Lek N, Carroll D, et al. Anogenital distance and penile length in infants with hypospadias or cryptorchidism: comparison with normative data. *Environ Health Perspect* 2014;122:207-11.
18. Cortes D, Fischer MB, Hildorf AE, et al. Anogenital distance in a cohort of 169 infant boys with uni- or bilateral cryptorchidism including 18 boys with vanishing testes. *Hum Reprod* 2024;39:689-97.
19. Andreasen SM, Gehrt L, Hagen CP, et al. Correlation of anogenital distance from childhood to age 9 years — a prospective population-based birth cohort — the Odense Child Cohort. *Hum Reprod Open* 2024;2024:hoae050.
20. Kiviranta P, Kuiri-Hänninen T, Saari A, Lamidi ML, Dunkel L, Sankilampi U. Transient postnatal gonadal activation and growth velocity in infancy. *Pediatrics* 2016;138(1):e20153561.
21. Varimo T, Hero M, Laitinen EM, et al. Childhood growth in boys with congenital hypogonadotropic hypogonadism. *Pediatr Res* 2016;79:705-9.
22. Davis SM, Kaar JL, Ringham BM, Hockett CW, Glueck DH, Dabelea D. Sex differences in infant body composition emerge in the first 5 months of life. *J Pediatr Endocrinol Metab* 2019;32:1235-9.
23. Becker M, Oehler K, Partsch C-J, et al.

- Hormonal 'minipuberty' influences the somatic development of boys but not of girls up to the age of 6 years. *Clin Endocrinol (Oxf)* 2015;83:694-701.
24. Lamminmäki A, Hines M, Kuiri-Hänninen T, Kilpeläinen L, Dunkel L, Sankilampi U. Testosterone measured in infancy predicts subsequent sex-typed behavior in boys and in girls. *Horm Behav* 2012;61:611-6.
 25. Pasterski V, Acerini CL, Dunger DB, et al. Postnatal penile growth concurrent with mini-puberty predicts later sex-typed play behavior: evidence for neurobehavioral effects of the postnatal androgen surge in typically developing boys. *Horm Behav* 2015;69:98-105.
 26. Kung KTF, Browne WV, Constantinescu M, Noorderhaven RM, Hines M. Early postnatal testosterone predicts sex-related differences in early expressive vocabulary. *Psychoneuroendocrinology* 2016;68:111-6.
 27. Wermke K, Hain J, Oehler K, Wermke P, Hesse V. Sex hormone influence on human infants' sound characteristics: melody in spontaneous crying. *Biol Lett* 2014;10:20140095.
 28. Forest MG, Cathiard AM, Bertrand JA. Evidence of testicular activity in early infancy. *J Clin Endocrinol Metab* 1973;37:148-51.
 29. Faiman C, Winter JSD. Sex differences in gonadotrophin concentrations in infancy. *Nature* 1971;232:130-1.
 30. Rohayem J, Alexander EC, Heger S, Nordenström A, Howard SR. Mini-puberty, physiological and disordered: consequences, and potential for therapeutic replacement. *Endocr Rev* 2024;45:460-92.
 31. Kuiri-Hänninen T, Seuri R, Tyrväinen E, et al. Increased activity of the hypothalamic-pituitary-testicular axis in infancy results in increased androgen action in premature boys. *J Clin Endocrinol Metab* 2011;96:98-105.
 32. Dhayat NA, Dick B, Frey BM, d'Uscio CH, Vogt B, Flück CE. Androgen biosynthesis during minipuberty favors the backdoor pathway over the classic pathway: insights into enzyme activities and steroid fluxes in healthy infants during the first year of life from the urinary steroid metabolome. *J Steroid Biochem Mol Biol* 2017;165:312-22.
 33. Andersson A-M, Toppaari J, Haavisto A-M, et al. Longitudinal reproductive hormone profiles in infants: peak of inhibin B levels in infant boys exceeds levels in adult men. *J Clin Endocrinol Metab* 1998;83:675-81.
 34. Bergadá I, Milani C, Bedecarrats P, et al. Time course of the serum gonadotropin surge, inhibins, and anti-Müllerian hormone in normal newborn males during the first month of life. *J Clin Endocrinol Metab* 2006;91:4092-8.
 35. Busch AS, Ljubicic ML, Upners EN, et al. Dynamic changes of reproductive hormones in male minipuberty: temporal dissociation of leydig and sertoli cell activity. *J Clin Endocrinol Metab* 2022;107:1560-8.
 36. Boisen KA, Kaleva M, Main KM, et al. Difference in prevalence of congenital cryptorchidism in infants between two Nordic countries. *Lancet* 2004;363:1264-9.
 37. Main KM, Toppaari J, Suomi A-M, et al. Larger testes and higher inhibin B levels in Finnish than in Danish newborn boys. *J Clin Endocrinol Metab* 2006;91:2732-7.
 38. Cortes D, Müller J, Skakkebaek NE. Proliferation of Sertoli cells during development of the human testis assessed by stereological methods. *Int J Androl* 1987;10:589-96.
 39. Ljubicic ML, Johannsen TH, Fischer MB, et al. Serum LH/FSH ratios in 87 infants with differences of sex development. *Endocr Connect* 2023;12(3):e220275.
 40. Aksglaede L, Sørensen K, Boas M, et al. Changes in anti-Müllerian hormone (AMH) throughout the life span: a population-based study of 1027 healthy males from birth (cord blood) to the age of 69 years. *J Clin Endocrinol Metab* 2010;95:5357-64.
 41. Scheutz Henriksen L, Holm Petersen J, Skakkebaek NE, et al. Serum testosterone levels in 3-month-old boys predict their semen quality as young adults. *J Clin Endocrinol Metab* 2022;107:1965-75.
 42. Cohen HL, Shapiro MA, Mandel FS, Shapiro ML. Normal ovaries in neonates and infants: a sonographic study of 77 patients 1 day to 24 months old. *AJR Am J Roentgenol* 1993;160:583-6.
 43. Kuiri-Hänninen T, Kallio S, Seuri R, et al. Postnatal developmental changes in the pituitary-ovarian axis in preterm and term infant girls. *J Clin Endocrinol Metab* 2011;96:3432-9.
 44. Fischer MB, Mola G, Rom AL, et al. Ovarian and uterine morphology in minipuberty: associations with reproductive hormones: a COPANA study of 302 girls. *J Clin Endocrinol Metab* 2025;110:1015-22.
 45. Ljubicic ML, Busch AS, Upners EN, et al. A biphasic pattern of reproductive hormones in healthy female infants: the COPENHAGEN minipuberty study. *J Clin Endocrinol Metab* 2022;107:2598-605.
 46. Kuiri-Hänninen T, Haanpää M, Turpeinen U, et al. Postnatal ovarian activation has effects in estrogen target tissues in infant girls. *J Clin Endocrinol Metab* 2013;98:4709-16.
 47. Ljubicic ML, Madsen A, Upners EN, et al. Longitudinal evaluation of breast tissue in healthy infants: prevalence and relation to reproductive hormones and growth factors. *Front Endocrinol (Lausanne)* 2022;13:1048660.
 48. Lee MM. Reproductive hormones in infant girls — a harbinger of adult reproductive function? *J Clin Endocrinol Metab* 2003;88:3513-4.
 49. Hagen CP, Fischer MB, Wohlfahrt-Veje C, et al. AMH concentrations in infancy and mid-childhood predict ovarian activity in adolescence: a long-term longitudinal study of healthy girls. *EClinicalMedicine* 2022;55:101742.
 50. Sarfati J, Bouvattier C, Bry-Gauillard H, Cartes A, Bouligand J, Young J. Kallmann syndrome with FGFR1 and KAL1 mutations detected during fetal life. *Orphanet J Rare Dis* 2015;10:71.
 51. Gueguen T, Martinier L, Castets S, et al. Inhibin B and AMH for diagnosis of hypogonadotropic hypogonadism in boys under 1 year of age: a case-control study. *J Clin Endocrinol Metab* 2025;110(12):e4119-e4128.
 52. Swee DS, Quinton R. Congenital hypogonadotropic hypogonadism: minipuberty and the case for neonatal diagnosis. *Front Endocrinol (Lausanne)* 2019;10:97.
 53. Thorup J, Petersen BL, Kvist K, Cortes D. Bilateral vanished testes diagnosed with a single blood sample showing very high gonadotropins (follicle-stimulating hormone and luteinizing hormone) and very low inhibin B. *Scand J Urol Nephrol* 2011;45:425-31.
 54. Jespersen K, Ljubicic ML, Johannsen TH, Christiansen P, Skakkebaek NE, Juul A. Distinguishing between hidden testes and anorchia: the role of endocrine evaluation in infancy and childhood. *Eur J Endocrinol* 2020;183:107-17.
 55. Cabrol S, Ross JL, Fennoy I, Bouvattier C, Roger M, Lahlou N. Assessment of Leydig and Sertoli cell functions in infants with nonmosaic Klinefelter syndrome: insulin-like peptide 3 levels are normal and positively correlated with LH levels. *J Clin Endocrinol Metab* 2011;96(4):E746-E753.
 56. Aksglaede L, Davis SM, Ross JL, Juul A. Minipuberty in Klinefelter syndrome: current status and future directions. *Am J Med Genet C Semin Med Genet* 2020;184:320-6.
 57. Hagen CP, Main KM, Kjaergaard S, Juul A. FSH, LH, inhibin B and estradiol levels in Turner syndrome depend on age and karyotype: longitudinal study of 70 Turner girls with or without spontaneous puberty. *Hum Reprod* 2010;25:3134-41.
 58. Bouvattier C, Carel J-C, Lecoindre C, et al. Postnatal changes of T, LH, and FSH in 46,XY infants with mutations in the AR gene. *J Clin Endocrinol Metab* 2002;87:29-32.
 59. Bin-Abbas B, Conte FA, Grumbach MM, Kaplan SL. Congenital hypogonadotropic hypogonadism and micropenis: effect of testosterone treatment on adult penile size why sex reversal is not indicated. *J Pediatr* 1999;134:579-83.

60. Samango-Sprouse CA, Sadeghin T, Mitchell FL, et al. Positive effects of short course androgen therapy on the neurodevelopmental outcome in boys with 47,XXY syndrome at 36 and 72 months of age. *Am J Med Genet A* 2013;161A:501-8.
61. Davis SM, Howell S, Janusz J, et al. Testosterone effects on short-term physical, hormonal, and neurodevelopmental outcomes (TESTO) in infants with 47,XXY. *J Clin Endocrinol Metab* 2025;110:3493-504.
62. Alexander EC, Faruqi D, Farquhar R, et al. Gonadotropins for pubertal induction in males with hypogonadotropic hypogonadism: systematic review and meta-analysis. *Eur J Endocrinol* 2024;190:S1-S11.
63. Pitteloud N, Hayes FJ, Dwyer A, Boepple PA, Lee H, Crowley WF Jr. Predictors of outcome of long-term GnRH therapy in men with idiopathic hypogonadotropic hypogonadism. *J Clin Endocrinol Metab* 2002;87:4128-36.
64. Rhys-Evans S, d'Aniello F, Alexander EC, et al. Gonadotropin therapy for mini-puberty induction in male infants with hypogonadotropic hypogonadism. *J Clin Endocrinol Metab* 2025;110(4):e921-e931.
65. Ren X, Wang Y, Liu Y, Gong C. Investigational treatment of congenital hypogonadotropic hypogonadism in infants. *Horm Metab Res* 2025;57:18-24.
66. Mesas-Aróstegui MA, Hita-Contreras F, López-Siguero JP. A therapeutic proposal for mini-puberty in male infants with hypogonadotropic hypogonadism: a retrospective case series. *J Clin Med* 2024;13:6983.
67. Stuckey BGA, Nolan JD, Hurley DM, Martin GB. Congenital hypogonadotropic hypogonadism, induction of mini-puberty, and future fertility. *Endocrinol Diabetes Metab Case Rep* 2023;2023:23-0038.
68. Young J, Xu C, Papadakis GE, et al. Clinical management of congenital hypogonadotropic hypogonadism. *Endocr Rev* 2019;40:669-710.

Copyright © 2026 Massachusetts Medical Society.

CLINICAL TRIAL REGISTRATION

The *Journal* requires investigators to register their clinical trials in a public trials registry. The members of the International Committee of Medical Journal Editors (ICMJE) will consider most reports of clinical trials for publication only if the trials have been registered. Current information on requirements and appropriate registries is available at www.icmje.org/about-icmje/faqs/.